

TG

Edits, deletions, and restrictions: Making Telegram's Content **Dynamics** Observable

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Piazzale Aldo Moro 5 · A.Y. 2025/26

19:24

Co-Advisor Alberto M. Mongardini joined the group

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The News

More than 100,000 people joined tonight's march.

19:24

The News

More than ~~100,000~~ 8,000 people joined tonight's march.

edited 19:23



48 comments •



History is lost. Catching this at scale is the main contribution.

19:24

message deleted.

detected on visit 4

48 comments



History is lost. Catching this at scale is the main contribution.

one pass per channel, then Done!

Prior corpora

```
Channel 0 scraped
Channel 1 scraped
  ↳ discussion group scraped
Channel 2 scraped
Done!
```

19:22

Kireev et al., 2025

We joined every channel and stayed connected; every new message was pushed to us live. We held **13 channels**.

19:23

To watch thousands of channels, you **cannot join them all**.

19:24

Prior corpora

Done!

```
Channel 0 scraped
Channel 1 scraped
  └ discussion group scraped
Channel 2 scraped
Revisiting...
```



19:23

A watched channel

Forwarded from a channel we hadn't seen yet

Discovery snowballs, one forward at a time.

19:24

Telegram

474.9 M messages. Re-reading all of it every visit
would take months.

19:23

One question, once per visit: how many messages
does this channel hold right now?

19:24



Pinned message

the count: how many messages the channel says it holds vs how many we hold

counts match

No messages were deleted, anywhere.
We re-read the **last N for edits**.

19:23

count short

The gap says **exactly how many** vanished.
We **walk backwards** and find them.
We check for edits along the way.

19:24

No exact timestamps, but 12,065 channels, not 13.

Corpus

Where the messages live

474.9 M messages

70% comment sections 333.2 M

30% broadcast channels 141.8 M



13 comments



The comment side outweighs all broadcasts, and gets the **same treatment**: revisited and diffed like every post.

19:24

TGDataset2

12,065 channels · 474.9 M messages

Public channels and their comment sections, revisited and diffed. Text and metadata: no media mirrored, except profile pictures.

Description

116.8 M messages with reactions

323 K polls

11,165 unique channel pictures

2.42 M detected deletions

36,016 diffed edits

18.7 M restriction flags

37,669 metadata snapshots

456 K channels referenced

Telegram

One message in three carries an `edit_date`.

19:23

Corpus

What did the edits we caught change?

share of edits we caught and diffed · 36,016 diffed edits

93% nothing: text byte-identical 33,660



7% the text actually changed 2,356

`edit_date` **over-counts** real rewrites **~10×**. Only the visit-to-visit diff finds them.

19:24

The News

~~40,000 attended.~~ 57,000 attended, per the organisers.

edited 19:23

Corpus

What kind of change was it?

2,356 genuine text rewrites

62% substantive: the claim moved 1,450

38% cosmetic: typos, phrasing 906

Rewrites quietly revise the record: the numbers walk up or down, the tone stays put.

19:24

message deleted · one of 2.42 M

Corpus

How toxic is what gets deleted?

mean toxicity · comments die 4× as often as posts · 61% go in purges

0.247 targeted deletions

0.187 blanket thread purges

0.128 what survives

Deleted content runs **1.62× more toxic** than what survives. Choice targets toxicity; purges don't.

18.7 M flagged messages · one channel, two phones

viewed from iOS

A watched channel



19:23

viewed from Android

Unfortunately, this channel can't be displayed on Telegram apps downloaded from the Google Play Store.

Same messages, different phone: 18.7 M carry a per-platform restriction flag, **96.8%** of them tagged **iOS** or **Android** specifically.

19:24

US/Israel-Iran · first strike 28 Feb 2026

Corpus

Which arm surges hardest?

× own pre-war baseline · Nov 2025 – Feb 2026

5.39× IDF · English

3.87× IDF · Hebrew

2.46× Mehr · English

1.90× Mehr · Farsi

IRNA repeats the shape (en 1.64× > fa 1.14×) · Tehran Times switched on from silence

Both wartime priorities: the international narrative.

19:24

wartime reactions, each scored positive or negative

Iranian cohort

Iran's official audience cheered the war on: reactions swung sharply positive, **+0.56** on the -1 to +1 scale.



2.41 M



0.49 M

19:22

Israeli cohort

Israel's barely moved: **+0.09**.



0.29 M



0.15 M

19:23

The English surge finds a real crowd: barely half of Iran's comments are in Farsi.

19:24

the scraper is still running

This thesis

System: a revisit-and-diff scraper. "

Corpus: 474.9 M messages, three change layers. "

Showcase: a live war through an institutional lens. "

19:23

limitations: detection-time samples · snowball discovery bias ·
corpus rates $\neq$ platform prevalence

Sustained running turns samples into **dynamics**.

19:24